

Understanding the Co-Design of Data-Driven Technologies in Urban Living Labs

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Abstract

Data-driven technologies such as artificial intelligence, blockchain, and real-time analytics are increasingly used to address complex societal and urban challenges. However, their sociotechnical complexity makes their design and governance contested. Urban living labs (ULLs) have gained traction as experimental environments where diverse stakeholders (municipalities, private sector organizations, researchers, and citizens) can tackle complex urban challenges together. However, there is a lack of scholarship on how the co-design of data-driven technologies unfolds within living labs.

This is why this PhD project investigates how data-driven technologies can be co-designed within urban living labs (ULLs). Drawing on interdisciplinary methods and literature from HCI, participatory design, and innovation management, the research aims to develop thick insights into the co-design of technology through ethnographic work conducted with case studies in the Dutch infrastructure sector. The work also aims to design and test small, situated interventions that can enhance specific pain points of the co-design process.

CCS Concepts

• **Human-centered computing** → **Computer supported cooperative work**; *Participatory design*; **Interaction design**; *Ethnographic studies*; **Collaborative and social computing theory, concepts and paradigms**; • **Applied computing** → **Computing in government**; • **Social and professional topics** → **Computer supported cooperative work**.

Keywords

Co-Design, Urban Living Labs, Co-Designing AI

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1 Introduction

To address complex urban challenges, cities are increasingly active in the development and adoption of data-driven technologies encompassing artificial intelligence, digital twins, Internet of Things (IoT) systems, and real-time analytics, among others. The term is deliberately broad: while these technologies differ in scope and maturity, they share common characteristics of technical complexity and reliance on large amounts of data, which make them difficult to co-design. Data-driven technologies are increasingly being adopted by cities around the world [14]. They can have great positive impacts on the efficiency of public government, but they also raise critical concerns about their transparency and accountability [2]. In many cases, they are introduced through top-down processes or vendor-driven solutions that do not account for the needs of diverse stakeholders [9]. This is why living labs (LLs) can be very relevant. LLs have gained traction as real-world environments where multiple stakeholders can engage in co-design and experimentation of innovative solutions. However, while LLs have been applied to domains such as sustainability and mobility, their potential to serve as a context for co-designing data-driven technologies remains underexplored.

Urban living labs (ULLs) can be particularly relevant for urban innovation. ULLs are multi-stakeholder innovation environments in which researchers, public sector actors, citizens, and private companies co-develop, test, and refine solutions to urban challenges in real-world settings [7, 26]. ULLs provide a setting for co-design and in situ experimentation, but they come with some challenges. In particular, the outcomes of living labs are often project-bound and do not translate into long-term institutional and organizational change. Many living labs conclude when project funding ends, leaving behind valuable prototypes but limited systemic change in the institutions meant to sustain them [11, 29]. This raises the question of how the co-design of technologies in living labs can become a driver of long-term transformation of cities, from an institutional perspective.

For these reasons, this doctoral project investigates how these data-driven technologies can be co-designed in ULLs. The core aim of this work is to understand how the process of co-designing data-driven technologies unfolds in urban living labs, and how it can become a driver of long-term impact, in terms of institutional and organizational change [23].

2 Background

2.1 Co-Designing Data-Driven Technologies in Cities

Cities are increasingly adopting data-driven technologies, including AI systems, digital twins, sensor infrastructures, and real-time analytics, to optimize planning, service delivery, and public-sector decision-making. These technologies bring several benefits, including improved efficiency and new forms of coordination; however, they also raise well-documented concerns related to accountability and the distribution of power in urban governance [2, 16, 17].

At the same time, co-design can be understood as a practice emphasizing the active involvement of stakeholders throughout all phases of the design process [27]. This study will adopt Sanders and Stappers' [22] definition of co-design as "the creativity of designers and people not trained in design working together in the design development process". Scholars have highlighted that co-design is not a single method, but a family of approaches and practices to foster the contribution of the participants [6, 27]. Co-design has been widely used in human-computer interaction [5]. Hence, co-design can become a method to engage communities and organizations in the creation of sociotechnical systems [18]. This is a way to address the limitations of traditional user-centered design, moving beyond designing for users to designing with them [25]. Existing studies in civic technologies have also proposed civic design frameworks that integrate public participation into the design and development of civic tools [21].

2.2 Urban Living Labs

Living labs emerged in the late 1990s and gained significant traction through the initiatives of the European Network of Living Labs. They are especially widespread in Europe, and to a lesser extent in East Asia and North America, and are commonly understood as Quadruple Helix innovation platforms involving collaboration among citizens, government, industry, and academia [3, 19]. The European Network of Living Labs defines living labs as "open innovation ecosystems in real-life test and experimentation environments based on a systematic user co-creation approach that integrates research and innovation activities in communities, placing citizens at the center of innovation, fostering co-creation and open innovation among the main actors of the Quadruple Helix Model" [10]. In this context, living labs have been widely positioned as promising governance arrangements for urban innovation. They are typically described as real-world, multi-stakeholder environments where public actors, private organizations, researchers, and citizens collaboratively develop and test novel solutions [4, 19]. Reflecting their highly interdisciplinary nature, living labs have also been widely studied within the field of Human-Computer Interaction. Alavi [1] describes four typologies of LLs in HCI, distinguishing between instrumented places (data-driven environments with sensor-based monitoring), visited places (spaces for periodic engagement and testing), lived-in places (everyday environments enabling continuous interaction), and innovation spaces (dynamic hubs for co-creation and experimentation) [1].

In the urban domain, urban living labs (ULLs) have been characterized as experimental spaces for sustainability-oriented innovation, enabling situated learning and iterative prototyping across institutional boundaries [8, 26, 30]. ULLs can thus be understood as sites of urban experimentation, where participation and design practices involve different stakeholders at the city level [7, 28].

2.3 Research Gap

We have seen that living labs are described as settings where diverse actors co-create solutions in real-world contexts [3, 19]. However, relatively little attention has been given to how living labs address the co-design of highly technical, data-driven solutions. While co-design is well established as a methodological approach within human-computer interaction [12, 22], there remains limited empirical and conceptual work on how co-design practices unfold in living labs when applied to technologies that are not easily understood or governed by non-experts [20, 30]. In this sense, the integration of data-driven technologies in living lab settings introduces specific challenges. First, many contemporary data-driven systems are complex and partially opaque [24]. Second, living labs frequently operate through project-based funding and time-bounded collaborations, which can limit long-term institutional embedding and reduce their capacity to produce systemic change [11, 29]. As a result, living labs may generate valuable prototypes and localized innovations while struggling to translate experimental outcomes into enduring organizational routines, governance arrangements, and policy commitments [1].

3 Research Questions

The overarching research question of this PhD project is: ***How can data-driven technologies be co-designed in urban living labs to drive long-term institutional impact?*** To address this question, this project investigates the practices, roles, and capacities that shape co-design, with the aim of explaining the conditions that can foster long-term transformation within living labs. To structure this exploration, the overarching research question is guided by four sub-research questions.

RQ1: *How has the co-design of data-driven technologies for transformative change in living labs been studied, and how can it be conceptualized?*

RQ2: *What practices of co-design emerge in urban living labs working on data-driven technologies?*

RQ3: *How do urban living labs design for (or neglect) post-project handover and institutional embedding?*

RQ4: *How can the co-design of data-driven technologies in living labs be supported to enable long-term impact and ownership?*

Taken together, these questions aim to explore the co-design of data-driven technologies in urban living labs and how it can lead to long-term change. The research moves from conceptual foundations (RQ1) to an empirical analysis of co-design processes (RQ2) and long-term impact (RQ3), toward actionable interventions (RQ4).

4 Planned Methodology

To address the research questions introduced in the previous section, the overall strategy of this PhD is primarily based on qualitative methods aimed at understanding the co-design of data-driven technologies in urban living labs. Ethnographic methods are particularly well-suited to the overarching research question, as they allow for an in-context understanding of how the co-design of data-driven technologies unfolds and under what conditions it succeeds.

This research is part of a Dutch innovation programme that brings together numerous public and private partners to foster social and technological innovation in the infrastructure sector. The programme includes urban living labs focused on sustainable infrastructure, where partners co-design and test practical innovations. These living labs constitute the empirical foundation of this doctoral research, serving as key case studies for examining how data-driven technologies are co-designed in real-world contexts.

4.1 Conceptual Foundations (RQ1)

The first component of this research focuses on establishing the theoretical and conceptual grounding of the dissertation. It will be primarily addressed through systematic and scoping reviews, case study analysis, and interviews with key stakeholders from ongoing or past living lab projects. It aims to situate the dissertation within existing research and to provide conceptual clarity, positioning the work within existing scholarship at the intersection of human–computer interaction, design studies, and living lab research, while identifying the conceptual gaps that motivate the inquiry. This phase involves reviewing and synthesizing relevant bodies of literature to clarify how key notions are currently understood.

4.2 Ethnography of Technology Co-Design in Urban Living Labs (RQ2 and RQ3)

The core empirical component of the project consists of an ethnographic study of co-design processes in Dutch living labs. This ethnographic research aims to document the situated practices of co-designing data-driven technologies as they unfold in selected ULLs. Through participant observation, informal conversations, and document analysis, the study will produce thick descriptions of these processes. This component contributes primarily to RQ2 and RQ3, as it will help make the dynamics of co-design practices for technology in urban living labs more visible and intelligible. The goal is to develop a practice-based understanding of the strategic use and development of technology co-design in living labs.

4.3 Interventions Situated in the Ethnographic Work

Building on insights from the ethnography, the project also aims to develop and test one or two small-scale design interventions in collaboration with stakeholders within the living lab context, addressing RQ4. These interventions will be modest in scope and will serve as a way to respond to challenges emerging from the fieldwork. They will be tested *in situ*, with the aim of supporting the co-design process. Possible forms include training activities

(e.g., to improve stakeholders' ability to engage with AI and data-driven technologies) or facilitation tools, such as frameworks that help living lab coordinators organize and run workshops. The final nature of the interventions will be decided iteratively based on findings from the ethnographic engagement. By designing and observing these situated interventions, the project aims to gain insights into how the co-design of data-driven technologies can be supported and strengthened in real-world contexts, directly addressing RQ4.

5 Main Challenges

5.1 Carrying Out an Ethnography of Co-Design in Living Labs

A key challenge has to do with setting up and conducting the ethnographic study at the core of the doctoral project. The longitudinal ethnography is situated within specific ongoing urban living labs from the Netherlands, with their idiosyncrasies and local dynamics, strongly linked to the Dutch infrastructure sector. Although this can provide deep and situated insights, it also means that the findings cannot be applied universally to other innovation contexts. Power asymmetries and competing accountabilities further complicate whose narratives and practices are reported ethnographically, while confidentiality and access constraints may reduce visibility into key decision-making processes.

Moreover, in order to design and test the interventions, I will need to be close enough to facilitate workshops and prepare interventions. Hence, to generate critical analysis, there will be a need for explicit reflexivity and positionality. A related limitation concerns the interventions conducted based on the insights obtained in the ethnography. These interventions are deliberately designed to support specific co-design practices, yet their scope is necessarily limited. They will likely take the form of events (such as workshops) or resources (such as booklets and methodologies), and their outcomes will be dependent on practical constraints such as timing and participant availability. Consequently, while these interventions could generate valuable insights into how co-design of data-driven technologies can be supported, they will not constitute definitive solutions.

5.2 Temporal Misalignment with Long-Term Change

One of the core limitations lies in the temporal boundedness of this project. The ethnography unfolds within the timeframe of a PhD project, which necessarily constrains what can be observed. Processes of institutional and organizational change typically develop over much longer timescales than those accessible within the research window. Demonstrating long-term transformation is methodologically difficult within the limited duration of a PhD program. At the same time, there is also an absence of well-established metrics that can help assess the transformative impact of living labs.

Hence, this dissertation will only be able to document emergent trajectories or report findings from urban living labs that concluded several years ago. However, the fact that there cannot be a long-term follow-up of the ULLs observed through the ethnographic

research may lead the main research question to only be partly addressed. Some contributions with respect to long-term impact will instead be addressed through case studies of living lab projects that concluded outside the timeframe of the main ethnographic study.

6 Intended Contributions

Building on the methodological approach outlined in Section 4, the intended contributions of this doctoral research are threefold, addressing theoretical, empirical, and methodological gaps:

- (1) **Theoretical Contribution:** The first goal is to advance theories of how different types of data-driven technologies can be co-designed within living labs to enable long-term change in cities. The project aims to contribute to conceptual discussions in HCI by exploring how processes of co-design can move beyond the production of localized prototypes to foster institutional and organizational transformation. This theoretical contribution aims to link design practice with participatory design concepts such as those of infrastructuring [15] and institutioning [13].
- (2) **Empirical Contribution:** The project will produce comparative evidence of the co-design of data-driven technologies in urban living labs, drawing from real-world implementations in the Dutch infrastructure sector. This will offer grounded knowledge of the core phenomenon, filling a pressing gap in the literature. Through ethnographic engagements, it will examine the key tensions occurring in real-life ULLs, highlighting the conditions that shape the continuity of co-designed outcomes. These findings aim to situate the practice of technology co-design within broader processes of policy-making and institutional learning.
- (3) **Methodological Contribution:** This research will develop a repertoire of co-design and ethnographic methods for studying and facilitating collaboration around data-driven technologies. The specific outcomes will be dependent on the ethnographic work, but they may introduce new approaches for building data literacy and cross-sector collaboration.

Beyond academic contributions, this doctoral research also aims to generate practical outputs aimed at supporting living lab practitioners, including designers and living lab coordinators, who facilitate innovation with emerging technologies. These outputs may include training materials, frameworks, handbooks, and manuals that translate HCI principles into actionable guidance for real-world practice. They will offer adaptable tools for structuring co-design processes, supporting data literacy, managing governance transitions, and embedding participatory routines within municipal organizations.

7 Conclusion

This doctoral research seeks to advance our understanding of how data-driven technologies can be co-designed within urban living labs to foster long-term changes in urban governance and organizational practice. By exploring the intersection of participatory design and digital innovation, this work positions living labs as crucial places for dialogue among multiple stakeholders to design and test new technologies. Through its contributions, this research

aspires to bridge a critical gap between short-term co-design experimentation with technology and institutional learning.

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